Fruit & Fly

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Why USDA’s PPQ matters

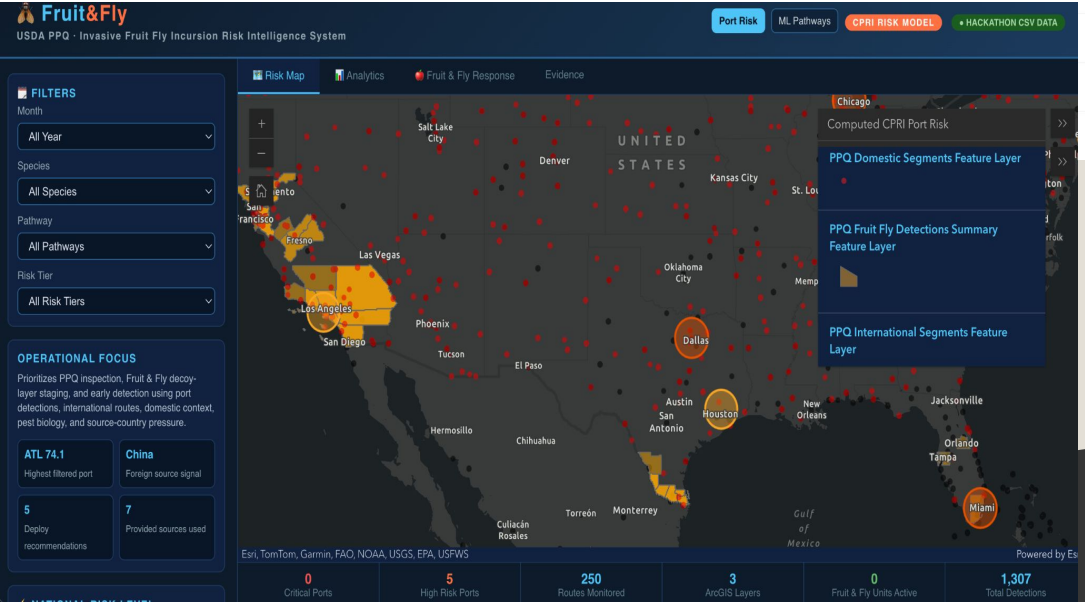
- Protection from invasive species and guarding our crops and forests
- USDA PPQ reports can provide insight on future risks and emergency responses before outbreaks cause major damage
- Optimizes surveillance pathways for fly migration

Male Attraction Techniques (MAT)

- Male wing vibrations attracted females
- Females are also attracted to male phermones

Data Integration and Pipeline

origin_country	us_port	passengers	month	year	fruit_imports	pest_status	fruit_fly_type	detections	detection_type	detected_fruit_fly_type
India	MIA	138832	2020-01	2020	0.0	Present	Oriental Fruit Fly (Bactrocera dorsalis)	7	Passenger baggage inspection	Mediterranean Fruit Fly (Cer capitata - Medfly)
Spain	DFW	138959	2020-01	2020	0.0	Absent	nan	5	Cargo shipment inspection	Mexican Fruit Fly (Anastrepl ludens)
Germany	MIA	168731	2020-03	2020	0.0	Absent	nan	6	Cargo shipment inspection	Mediterranean Fruit Fly (Cer capitata - Medfly)
Brazil	LAX	205062	2020-06	2020	0.0	Present	Mediterranean Fruit Fly (Ceratitis capitata - Medfly)	11	Passenger baggage inspection	Oriental Fruit Fly (Bactrocer dorsalis)
Brazil	MIA	382998	2020-06	2020	0.0	Present	Mediterranean Fruit Fly (Ceratitis capitata - Medfly)	10	Passenger baggage inspection	Mexican Fruit Fly (Anastrepl ludens)
Canada	IAH	281880	2020-06	2020	0.0	Absent	nan	2	Passenger baggage inspection	Mediterranean Fruit Fly (Cer capitata - Medfly)
							nan	11	Passenger baggage inspection	Oriental Fruit Fly (Bactrocer dorsalis)
							None	3	Passenger baggage inspection	Mediterranean Fruit Fly (Cer capitata - Medfly)
							nan	5	Cargo shipment inspection	Mediterranean Fruit Fly (Cer capitata - Medfly)



CSV

Feature Layer

Risk weights for ML model

Risk Analysis Model

Fruit&Fly ML Pathway Forecast

Predicted origin country to U.S. port entry pressure with destination temperature suitability, remaining 2026

Main Dashboard | ML Pathways

PPQ PATHWAY INTELLIGENCE

Ranks foreign-origin fruit fly entry pathways by predicted detections, host movement pressure, pest status, U.S. port temperature suitability, and Fruit & Fly decoy-layer response priority.

- **<40F masked** — Unsuitable for survival
- **40-45F marginal** — Lower confidence habitat
- **45F+ suitable** — Fruit & Fly response window
- **PPQ focus** — Early detection at ports

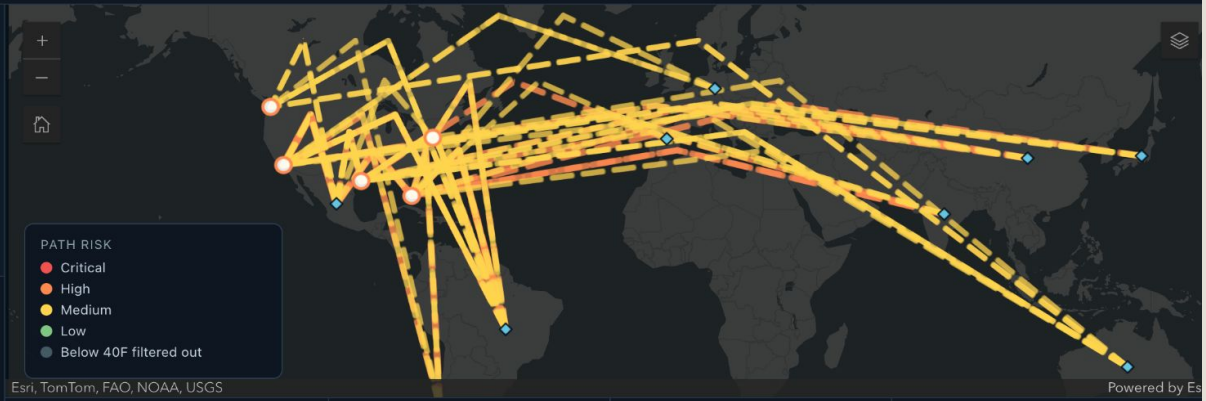
All risk tiers | All forecast months | All U.S. ports | All actionable temperature cla

Filter origin country

870 FILTERED PATHS	27 HIGH/CRITICAL
75.2% TOP CLIMATE SCORE	169.6 ADJUSTED DETECTIONS

RESPONSE BRIEF

Mexico → LAX



FRUIT & FLY HOTSPOTS

#1	LAX — Mexico · May · 58F	75.2%
#2	MIA — Mexico · Jun · 77F	74.4%
#3	IAH — Brazil · Jul · 76F	67.4%
#4	JFK — Brazil · Sep · 62F	59%
#5	SEA — Brazil · Sep · 53F	52.8%

ORIGIN COUNTRY PRESSURE

Mexico	23.6
Brazil	20.5
China	14
Chile	13
Australia	13
India	12.6
Spain	11.7
Japan	10.6

PORT PRESSURE

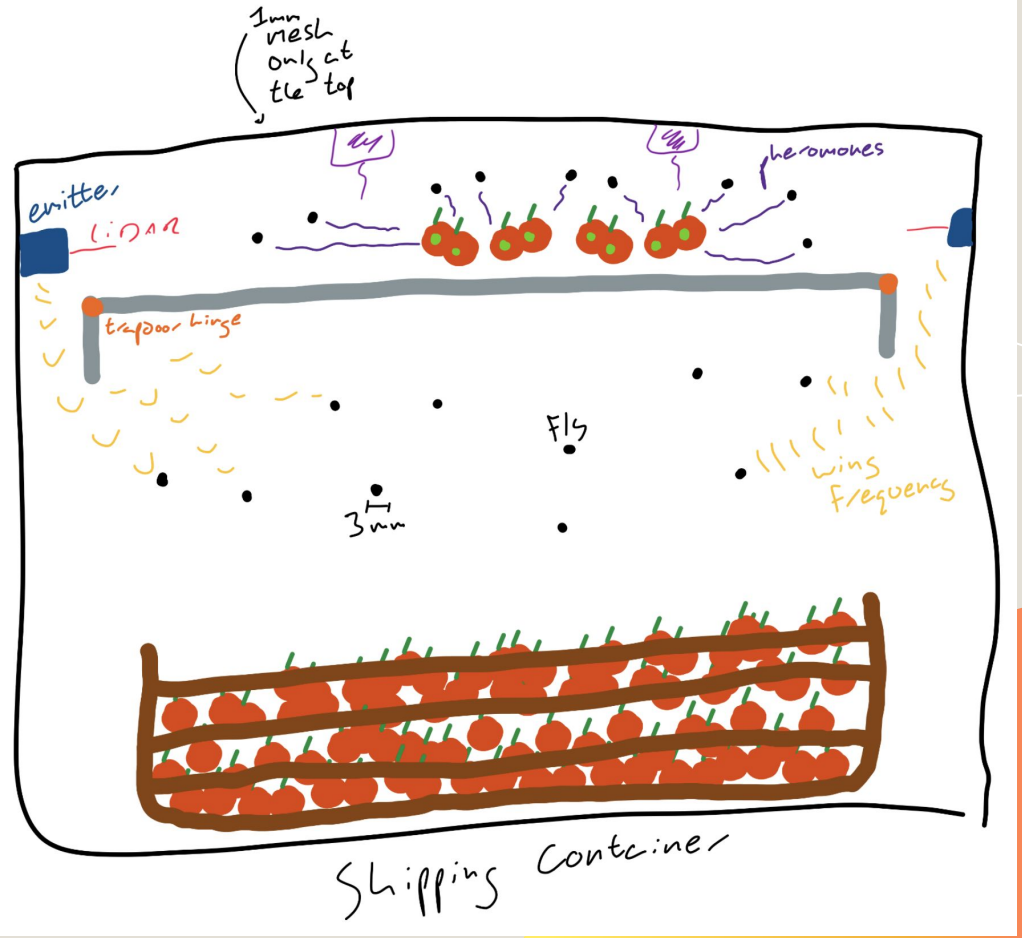
LAX	37.2
MIA	33.4
IAH	28.5
SEA	25.4
JFK	16.5
DFW	10.4
ATL	10.1
ORD	8.1

EVIDENCE FRAMEWORK

Geospatial	Origin-country to U.S.-port arcs, port response hotspots, and filterable destination layers.
Prediction	27 high/critical pathways from 15 foreign sources into 8 U.S. ports.
Biology	90 below-40F pathway-months masked; 45 40-45F pathway-months kept as marginal.

Our Solution:

“Fruit & Fly”



How We Can Help

- Our solution could potentially minimize the arrival of invasive species to the U.S. such as the Mediterranean, Mexican, and Oriental fruit fly

- Further preventing of economic harm done by invasive species

- Allow for resource allocation elsewhere

Works Cited

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